APIs in SAP Business Network Freight Collaboration

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SAP Business Network Freight Collaboration | LBN

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APIs in SAP Business Network Freight Collaboration

Shippers and carriers can use the APIs in SAP Business Network Freight Collaboration, to simplify freight tendering, subcontracting, and dock appointment scheduling processes.

Code Lists

SAP Business Network for Logistics has defined its own code values for the various attributes used within SAP Business Network for Logistics. These codes are used during B2B communication so using APIs in SAP Business Network for Logistics. For more information on codes supported by business network, see Code Lists.

API Payload Limits

To prevent APIs from being overloaded by the payload size, we have introduced API payload limits to regulate the API usage per tenant. The use by client applications or background job threads that make multiple high-volume calls continuously to extract or import data, or update can cause performance issues.

The following list provides the overview of the proposed maximum number of items per document for various APIs:

API Name	Technical Name	Maximum Number of Items per Document
Provider API to Create Freight Request for Quotation	`TransportationOrderQuotationCreateRequest_Out`	700
Provider API to Cancel Freight Request for Quotation	`TransportationOrderQuotationCancellationRequest_Out`	700
Provider API to Notify Freight Quotation	`TransportationOrderQuotationNotification_Out`	700
Provider API to Create Freight Order for Subcontracting	`TransportationOrderRequest_Out`	700
Provider API to Cancel Freight Order for Subcontracting	`TransportationOrderCancellationRequest_Out`	700
Provider API to Create Freight Booking	`TransportationOrderBookingRequest_Out`	700
Provider API to Cancel Freight Booking	`TransportationOrderBookingCancellationRequest_Out`	700
Provider API for Freight Booking Waybill Notification	`TransportationOrderBookingWaybillNotification_Out`	700
Provider API to Create Freight Order for Settlement	`TransportationOrderRequest_Out`	700
Provider API to Confirm Freight Order Dispute Charges	`TransportationOrderChargeElementConfirmation_Out`	700

If a document reaches the maximum item count, the system rejects the document.

Rate Limiting

API requests for SAP Business Network Freight Collaboration are subject to rate limits.

Rate limits restrict the number of requests made by a caller within a defined time period. Batch jobs in external systems can cause a sudden high load in SAP Business Network Freight Collaboration which may not comply with the fair usage policy. Hence, it is recommended that you use higher rate limits for critical time bound processes, which are available only for a certain period of time. For non-critical processes, you can phase the requests through the day based on the rate limits.

When you exceed the rate limit, your request may be rejected by the network.

The rate limits are as follows:

- The rate limit per tenant is 1 request per second.
- The rate limit per tenant is 5 requests per second for 4 hours in a day beyond which the rate limit switches back to 1 request per second.

This rate limit is applicable across all message types in SAP Business Network Freight Collaboration.

Carrier APIs (REST APIs)

As a carrier, you can use the APIs in SAP Business Network Freight Collaboration to perform your daily tasks in the areas of freight tendering, freight subcontracting, freight booking, freight tracking, and freight settlement.

Note that when you pass a value for a field using the REST APIs in SAP Business Network Freight Collaboration, you must ensure that the length of the value adheres to the maximum length specified for the particular field in the shipper's transportation management system.

Freight Contracting

The freight contracting process comprises of the freight subcontracting and tendering processes.

Subcontracting

You can participate in freight tendering when an ordering party sends you a freight request for quotation (freight RFQ).

Your options for responding to the received freight RFQ are determined by the tendering setup implemented by the ordering party.

You can perform the following activities using the APIs for freight tendering:

- You can use the Provider API for Freight Tendering to accept or reject a freight request for quotation.
- You can use the Consumer API for Freight Tendering to create or cancel a freight request for quotation in an external system.
- You can use the Consumer API for Freight Tendering to award or reject a freight quotation in an external system.

More Information

Freight Tendering.

Subcontracting

As a subsequent step to tendering process, if an ordering party wants to subcontract the transportation of goods to you, you can use the freight subcontracting APIs.

You can perform the following activities using the APIs for freight subcontracting:

- You can use the Provider API for Freight Subcontracting to confirm or reject a freight order for subcontracting.
- You can use the Consumer API for Freight Subcontracting to create, update, or cancel a freight order for subcontracting in an external system.
- You can use the Consumer API for Freight Subcontracting to award a freight order for subcontracting in an external system.

More Information

Subcontracting

Consumer API for Freight Tendering

As a carrier, you can use this consumer API to manage freight requests for quotation and freight quotations as part of the freight tendering process when integrating SAP Business Network for Logistics with other systems.

Description

You can use this outbound API to perform the following:

- **Create Freight Request for Quotation**

  Enables you to create a freight request for quotation in an external system.

- **Cancel Freight Request for Quotation**

  Enables you to cancel a freight request for quotation in an external system.

- **Freight Quotation**

  Enables you to award or reject a freight quotation in an external system.

Additional Information

- If the **Currency Change Enabled** indicator is enabled by the ordering party, you can change the price proposal currency in the freight request for quotation. If it is disabled, then you cannot modify the price proposal currency.
- For the `TransportModeCode` attribute, when the ordering party sends a value from their external transportation system, the respective value that you receieve in SAP Business Network for Logistics might be different as the static mapping of values between these two systems are different.

Transport Mode Codes

SAP Transportation Management System	SAP Business Network for Logistics System
03 - Ocean	01 - Ocean
01 - Road	03 - Road

SAP Transportation Management System	SAP Business Network for Logistics System
05 - Air	04 - Air

Provider API for Freight Tendering

As a carrier, you can use this provider API to manage freight requests for quotation and freight quotations as part of the freight tendering process when integrating SAP Business Network for Logistics with other systems.

Enables you to create a freight quotation for a freight request for quotation in SAP Business Network for Logistics.

Description

You can use this inbound API to accept or reject a freight request for quotation.

When accepting, you can perform the following if permitted by the ordering party:

- Propose an alternative price

  The alternative price is the amount for which you agree to execute the transportation.

- Propose alternative loading and unloading times for transportation stages

The alternative times are the points in time for which you agree to accept the freight request for quotation.

When rejecting, you can provide a reason code.

Consumer API for Freight Subcontracting

As a carrier, you can use this consumer API to manage freight orders for subcontracting process when integrating SAP Business Network for Logistics with other systems.

Description

You can use this outbound API to perform the following:

- **Create Freight Order for Subcontracting**

  Enables you to create a freight order for subcontracting in an external system. The system calls this API when an ordering party creates a freight order for subcontracting.

- **Update Freight Order for Subcontracting**

  Enables you to update a freight order for subcontracting in an external system. The system calls this API when an ordering party updates a freight order for subcontracting.

- **Cancel Freight Order for Subcontracting**

  Enables you to cancel a freight order for subcontracting in an external system. The system calls this API when an ordering party cancels a freight order for subcontracting.

- **Award Freight Order for Subcontracting**

Enables you to award a freight order for subcontracting in an external system. The system calls this API when an ordering party awards a freight order for subcontracting to you.

Connection Information

Technical Name	Field
Company Type	Carrier
Version	1.0.0
Format	JSON
Base Document Type	Type of freight document supported: • 1122 – Transportation Order • 610 – Transportation Booking Order

You can download the YAML file from the Consumer API for Freight Subcontracting in SAP API Hub.

Data Dictionary

In the SAP API Hub, in the **Consumer API for Freight Subcontracting**, you can view the field details in the **Model View**. In the **API Reference** section, you can view the example values.

In the downloaded YAML file you can view the following details of the fields in the API interface:

- Field name
- Type
- Description
- Max length
- Required or not
- Example

Relevant Codes

SAP Business Network for Logistics has defined its own code values for the various attributes used within business network. These codes are used during B2B (business-to-business) communication so using APIs in business network. For more details on the list of codes supported in SAP Business Network for Logistics, see Code Lists.

For consumer API for freight booking, the following code lists are relevant:

- Container Types
- Currency Codes
- Document Reference Type Codes
- Exchange Rate Types
- Incoterm Descriptions

- Item Category Code
- Language Code
- Package Material Type Descriptions
- Shipping Type Descriptions
- Time Zone
- Transport Means Description Code
- Unit of Measure Length
- Unit of Measure Quantity
- Unit of Measure Temperature
- Unit of Measure Volume
- Unit of Measure Weight

Additional Information

- If the **Currency Change Enabled** indicator is enabled by the ordering party, you can change the price proposal currency in the freight order for direct tendering scenario. If it is disabled, then you cannot modify the price proposal currency.
- For the `TransportModeCode` attribute, when the ordering party sends a value from their external transportation system, the respective value that you receieve in SAP Business Network for Logistics might be different as the static mapping of values between these two systems are different.

Transport Mode Codes

SAP Transportation Management System	SAP Business Network for Logistics System
03 - Ocean	01 - Ocean
01 - Road	03 - Road
05 - Air	04 - Air

Possible Errors

- If the carrier note is updated directly by shipper in their **SAP Transportation Management** system, you will not receive that carrier note from shipper tenant.

Provider API for Freight Subcontracting

As a carrier, you can use this provider API to manage freight orders for subcontracting process when integrating SAP Business Network for Logistics with other systems.

The ordering party sends a freight order from their transportation system to SAP Business Network for Logistics. If you have integrated to the freight order confirmation process using the **Consumer API for Freight Subcontracting**, then it would be available in your backend system. You can use this inbound API to create a response for a freight order for subcontracting in SAP Business Network for Logistics. With this functionality, you can confirm or reject a freight order for subcontracting.

When confirming, you can perform the following if permitted by the ordering party:

- Propose an alternative price (in case of direct tendering)

  The alternative price is the amount for which you agree to execute the transportation.

- Provide driver names and vehicle details. Don't provide these information for consignment orders.
- Provide notes and attachment documents
- Provide container identification numbers and reference document number

When rejecting, you can provide a reason code.

Description

You can use this inbound API to create a response for a freight order (or direct tendering) in SAP Business Network for Logistics. With this functionality, you can confirm or reject a freight order.

The ordering party sends a freight order for the reservation of freight space from their transportation system to SAP Business Network for Logistics. If you have integrated the freight order confirmation process using the **Consumer API for Freight Subcontracting**, then it would be available in your backend system. You can respond to the freight order by either confirming or rejecting using the **Provider API for Freight Subcontracting**.

You can submit a payload with confirmation soon after you receive a freight order. Later, you can update the details and resend your confirmation. Additionally, you can view the road freight orders in **Freight Order for Confirmation** app in SAP Business Network for Logistics.

Whenever there is direct tendering involved, you can see the freight orders using the same app with remaining time to respond back. You must ensure that you respond before the remaining time. Every update for direct tendering creates a new quotation in the shipper tenant in the external transportation system.

Connection Information

Technical Name	Field
Company Type	Carrier
Version	1.0.0
Format	JSON
Base Document Type	Type of freight document supported: • 1122 – Transportation Order • 610 – Transportation Booking Order

You can download the YAML file from the Provider API for Freight Subcontracting in SAP API Hub.

Data Dictionary

In the SAP API Hub, in the **Consumer API for Freight Subcontracting**, you can view the field details in the **Model View**. In the **API Reference** section, you can view the example values.

In the downloaded YAML file you can view the following details of the fields in the API interface:

- Field name
- Type
- Description
- Max length
- Required or not
- Example

Relevant Codes

SAP Business Network for Logistics has defined its own code values for the various attributes used within business network. These codes are used during B2B (business-to-business) communication so using APIs in business network. For more details on the list of codes supported in SAP Business Network for Logistics, see Code Lists.

For provider API for freight subcontracting, the following code lists are relevant:

- **Document Reference Type Codes**
- **Currency Codes**

Possible Errors

- The shipper can maintain the necessary settings to disable you from changing the price, currency, or both. In this scenario, if you respond back to the shipper with these values, then the update fails.

Freight Booking

Using the freight booking APIs, you can review and confirm freight bookings that ordering parties ask you to execute.

You can perform the following activities using the APIs for freight booking:

- You can use the Provider API for Freight Booking to confirm or reject a freight booking.
- You can use the Consumer API for Freight Booking to create or cancel a freight booking request in an external system.
- You can use the Consumer API for Freight Booking to create or update a freight booking waybill information in an external system.

More Information

Freight Bookings for Confirmation

Consumer API for Freight Booking

As a carrier, you can use this consumer outbound API to manage ocean or air freight booking process when integrating SAP Business Network for Logistics with other systems.

Description

You can use this outbound API to perform the following:

- **Create Freight Booking Request**

  Enables you to create or update a freight booking request in an external system. The system calls this API when an ordering party creates or updates a freight booking request.

- **Cancel Freight Booking Request**

  Enables you to cancel a freight booking request in an external system. The system calls this API when an ordering party cancels a freight booking request.

- **Create Freight Booking Waybill Notification**

  Enables you to create or update a freight booking waybill information in an external system. The system calls this API when an ordering party creates or updates a freight booking waybill notification.

Connection Information

Technical Name	Field
Company Type	Carrier
Version	1.0.0
Format	JSON
Base Document Type	Type of freight document supported: • 1122 – Transportation Order

You can download the YAML file from the Consumer API for Freight Booking in SAP API Hub.

Data Dictionary

In the SAP API Hub, in the **Consumer API for Freight Booking**, you can view the field details in the **Model View**. In the **API Reference** section, you can view the example values.

In the downloaded YAML file you can view the following details of the fields in the API interface:

- Field name
- Type
- Description
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Relevant Codes

SAP Business Network for Logistics has defined its own code values for the various attributes used within business network. These codes are used during B2B (business-to-business) communication so using APIs in business network. For more details on the list of codes supported in SAP Business Network for Logistics, see Code Lists.